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CHENNAI
ACADEMY OF
ARCHITECTURE AND
DESIGN

POONAMALLEE CITY CAMPUS,
CHENNAI.

NATA NEW FORMAT

GUIDED LEARNING RESOURCES

(PREPARED BY EXPERT FACULTY IN ARCHITECTURE EDUCATION)



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DESIGN SENSITIVITY

Ability to observe record and analyze, people, space, product, environment. Critical thinking, reasoning and ability to identify the subtle communications.

DESIGN SENSITIVITY

DESIGN SENSITIVITY

DESIGN SENSITIVITY – INTRODUCTION

In our daily lives, we come across many observations related to people, products, spaces and the environment. We develop a sensitivity based on observation and interactions which makes us judge whether something is apt, comfortable, appealing or vice-versa. These interpretations are based on many standards we are habituated with. The degree to which a person responds to relevant aesthetic stimuli with reference to design that is consistent and appropriate with established, external aesthetic value standards may be referred to as aesthetic/design sensitivity. Design sensitivity is the care with which a design connects to its specific context.

Design sensitivity tests will evaluate a candidate's **perception, imagination and observation; creativity and communication; and architectural awareness**. They will also test our observation skills regarding basic ideas of design in everyday life, in the areas of **perspective, shapes, color theory, elements and principles of design, and the history of art**. These tests will gauge our ability to critically think and reason in given situations and also check if we can identify subtle communications.

The questions from this part of the syllabus would focus on:

- Applying attention to certain details, analysing, reasoning, classifying, inferring and predicting.
- Ability to discern subtle differences in visual properties and aesthetic outcomes
- Awareness of persons, places, Buildings, Materials, Objects, Texture related to Architecture and built environment.
- Visualizing three dimensional objects from two dimensional drawings.
- Visualizing different sides of three dimensional objects

When you are taking design sensitivity tests, you must deduce the outcomes based on the information presented in relation to observations and general knowledge.

TOPICS TO BE COVERED FOR A DESIGN SENSITIVITY TEST

The types of Questions may be based on the following areas:

1. COLOUR THEORY
2. SHAPE THEORY
3. DESIGN THEORY – ELEMENTS AND PRINCIPLES OF DESIGN
4. PERSPECTIVE
5. LOGOS
6. TYPOGRAPHY
7. MATERIALS IN DESIGN
8. ARCHITECTURAL AWARENESS

HOW TO USE THE READING MATERIAL FOR THIS SECTION

The reading material has been collated by interpreting the syllabus given in the NATA Brochure 2024. The student can use the material to get an idea about the areas based on which the questions may be framed.

It is suggested that extra reading be done on given content and related topics. References and links to further reading have been provided at the bottom of each page.

DESIGN SENSITIVITY

01. COLOUR THEORY

INTRODUCTION

Color is one of the basic elements of visual perception. We perceive colors first. Color is also a characteristic feature of architecture and urban space. We notice the color of the building earlier than its shape. Color is an extremely evocative medium with the power to elicit immediate and clear reactions from the observer. Its use in architecture and the built environment radically affects the perception of space and architectural form. Color is therefore of prime importance as an integral element of architectural design.

COLOUR THEORY

Color is our visual perception of different wavelengths of light. Color Theory describes how different colors relate to each other, and how they look when they are combined into many color schemes in design. Color theories create a logical structure for color. An offshoot of color theory is color psychology, which explores colors and emotions perceived by people within the context of their cultures.

Color Terminology:

The **color spectrum** refers to all the hues in visible light that are detectable to the human eye.

Hue derives from the Middle English “hewe,” which means “color.”

Chroma, or saturation, refers to the intensity or purity of colors seen on the color wheel above. A pure hue is more saturated than its tint or tone.

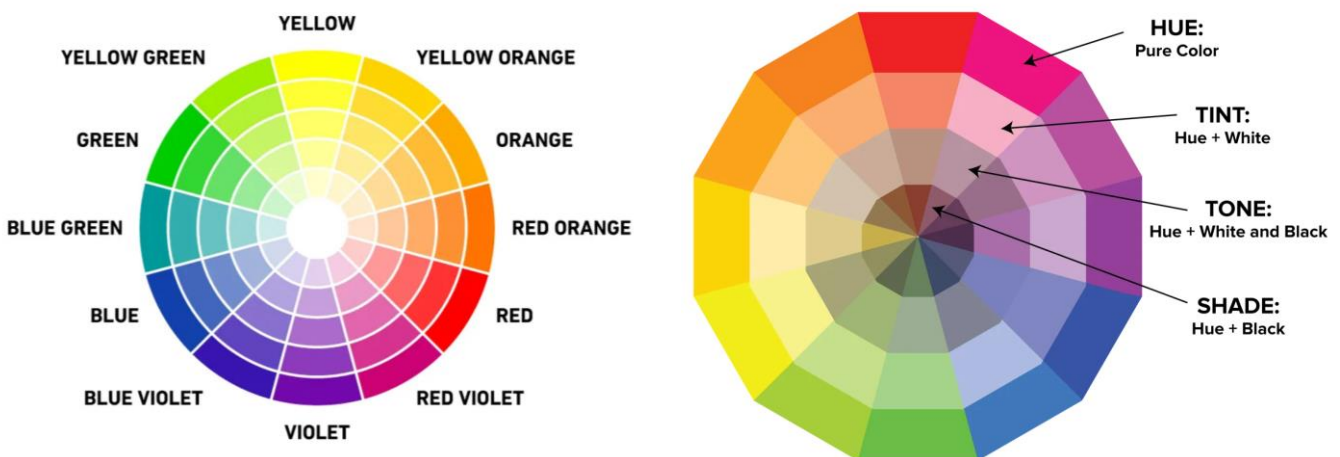
Tints are achieved by incorporating elements of white to brighten and desaturate a single hue. Tints of a color are considered more subdued or duller than their saturated counterparts.

Tones are achieved by adding gray to a hue, dulling the overall chroma.

Shades are achieved by adding portions of black to a single hue, creating a darker hue.

Value refers to the general lightness or darkness of a color. A lighter hue has more value than a darker hue.

Temperature refers to the perceived warmth or coolness of a hue. Reds, oranges, and yellows are associated with warmth due to their associations with warm objects in nature, such as fire or sunshine. The other side of the color spectrum blues, in particular, are considered cool colors because they're seen in ice, water, etc.



References:

<https://blog.hubspot.com/marketing/color-theory-design>

<https://www.shutterstock.com/blog/complete-guide-color-in-design>

DESIGN SENSITIVITY

02. SHAPE THEORY

INTRODUCTION

Design is a visual language that sends messages to your target audience and even the shapes you use contribute to that message. In fact, shape and form have been carrying meaning for centuries! Visual language was used for storytelling, guidance, and learning for early humans before spoken language evolved, and we continue to understand messages relayed through visual communication to this day.

Three Categories of Shape in Design

Shapes in design are categorized into three main types: geometric shapes, natural (or organic) shapes, and abstract shapes. Each of these three basic shapes groups and their meanings bring something very different to a design.

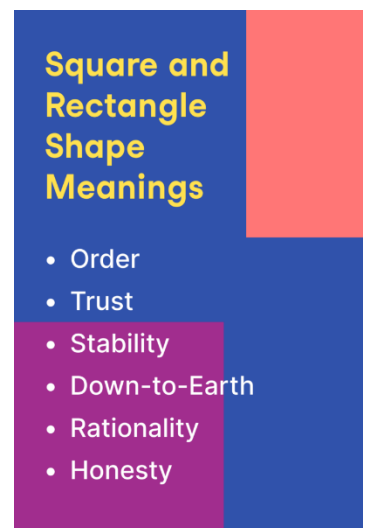
Geometric Shapes - They include squares, rectangles, triangles, diamonds, circles, ovals etc. These shapes are easily identified and have been given names. The shapes with straight lines and angles usually symbolise structure and order, while the shapes with curves are softer and represent connection and community.

Circle: The circle is a centralized figure that is normally stable and self-centered in its environment. Placing an element along its circumference or angular forms can induce circular motion

Triangles: A triangle is stable when sitting on its base, and unstable when it is on its side or inverted. Depending on how it is positioned, a triangle represents stability and instability, an inherent tension throughout the design. It suggests action, movement, and in some cases aggression.

Square: Squares symbolize order, trust, stability, and being grounded. One of the most familiar geometric shapes in design, squares and rectangles are used to organize and define a design, adding comforting structure along the way. These shapes suggest rational, practical and conformity.

The square is a symmetrical figure having two equal and perpendicular axes. All other rectangles can be considered as variations of the square varying in height or width. Similar to a triangle, the square is stable when it rests on one of its sides and tends to fall when it stands on one of its vertices.



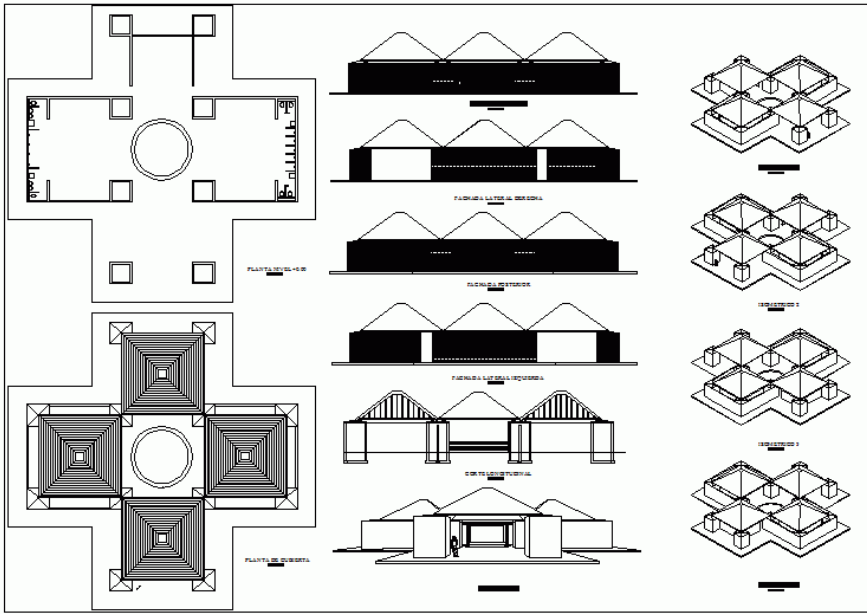
Reference:

<https://thearchspace.com/an-important-element-of-design-form-in-architecture/>

<https://www.tailwindapp.com/blog/the-meaning-of-shapes-in-design>

DESIGN SENSITIVITY

02. SHAPE THEORY contd...



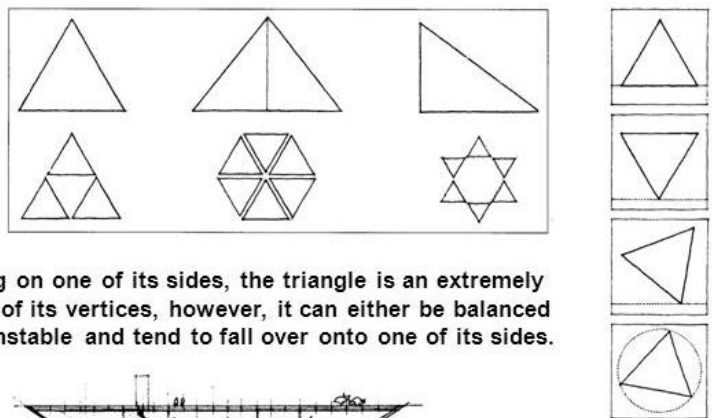
Bathhouse, Jewish Community Center, Trenton, New Jersey.;



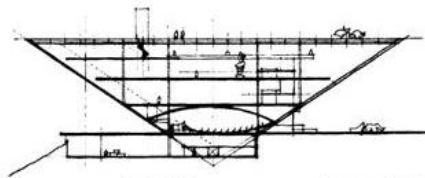
The Epidauros Amphitheatre in Greece

PRIMARY SHAPES

2-TRIANGLE

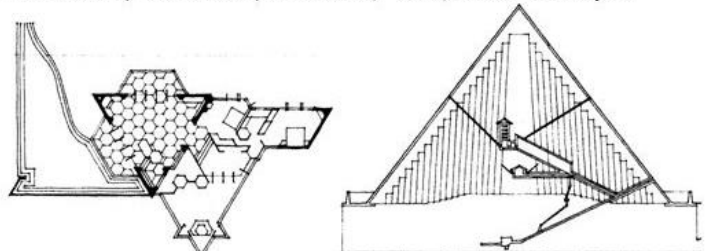


The triangle signifies stability. When resting on one of its sides, the triangle is an extremely stable figure. When tipped to stand on one of its vertices, however, it can either be balanced in a precarious state of equilibrium or be unstable and tend to fall over onto one of its sides.



Modern Art Museum, Caracas, Venezuela, 1955, Oscar Niemeyer.

Vigo Sundt House, Madison, Wisconsin, 1942, Frank Lloyd Wright.



Great Pyramid of Cheops at Giza, Egypt, c. 2500 B.C.

DESIGN SENSITIVITY

02. SHAPE THEORY contd...



The Guggenheim Museum, Bilbao, Spain.



The Heydar Aliyev Center, Azerbaijan, Baku

Organic Shapes

Organic, or natural, shapes in design are associated with nature. In other words, they often **look similar to things found in nature like plants, animals, and rocks**. They tend to be characterized by curving lines and a lack of symmetry. Unlike geometric shapes, organic shapes look far less solid and precise without stiff corners or straight lines.

Organic shapes have meaning in design that may differ a little bit depending on the particular shape you're using.

For example, if you use a wave shape in your design, your viewer will probably associate it with **freedom and flow**. If you use a leaf or a flower shape, your design might evoke the **feeling of nature and delicacy**.

Abstract Shapes

When you hear the term abstract shapes, you might think of abstract art and curving shapes that don't really look like anything in particular. You're probably thinking of organic shapes, which we covered above. It's a mistake many people make, because abstract shapes — and even abstract art — are all about interpretation. **It doesn't actually matter if the abstract shape looks more organic or geometric; it's more about the process and the context behind it.**

A good example of an abstract shape is an icon. Although abstract shapes often have geometric lines in them, they are abstract because they symbolize or reference something else that depends entirely on the context of their use.



The Royal Ontario Museum in Toronto, Canada



House of Dior - Seoul, Korea

Reference: <https://archi-monarch.com/geometry-in-architecture/>

DESIGN SENSITIVITY

03. DESIGN THEORY

Design Theory focuses on understanding the principles of creativity and design.

Design theory is a system of ideas that explains how and why design works. It includes definitions of elements and principles of creating design. The number of design elements and principles is not fixed and can vary depending on the context.

Design Elements are the simplest pieces within a design.

Design Principles are the ways elements or pieces are put together.

Design theory is why you are placing the elements or pieces of the design together in different ways.

Design theory is why one piece of design works and another doesn't.

Design Elements

The basic general elements of architectural design are...

Point – Points serve as focal markers or pivotal elements or significant junctions, anchoring attention within a design.

Line – Lines define boundaries and movement and guide the eye through compositions or space.

Shape – Shapes define the outlines of elements within a composition, shaping visual structure.

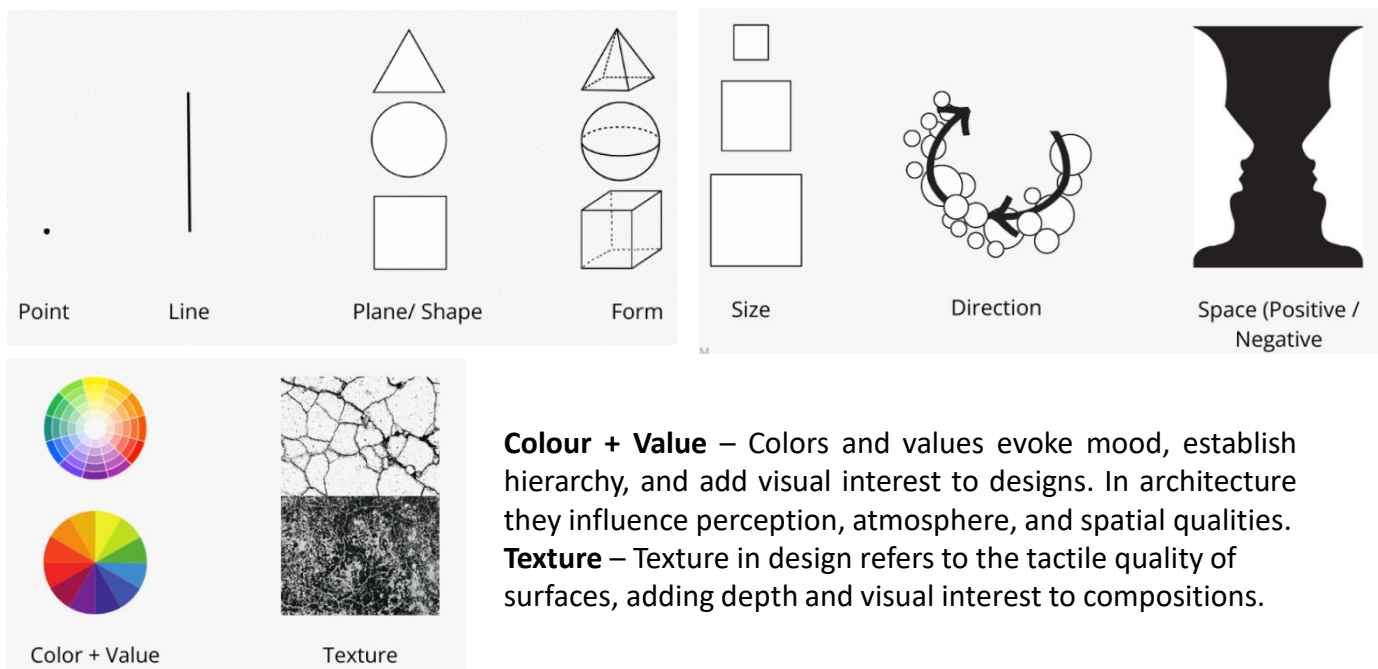
Form – Form refers to the three-dimensional aspect of design, encompassing mass, volume, and spatial arrangement.

Size – Size denotes the physical dimensions of elements, influencing scale and proportion within architectural compositions.

Direction – Refers to the orientation and alignment of elements. It encompasses the lines, shapes, or forms that guide the viewer's eye through the design and create a sense of movement or flow.

Direction can also be used as a design principle.

Space (Positive + Negative) – In design, positive space refers to the occupied areas within a composition, while negative space represents the unoccupied or empty areas, both contributing to the overall balance and harmony of the design.



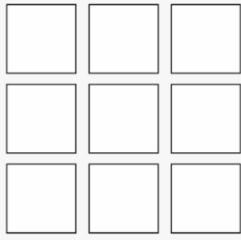
Colour + Value – Colors and values evoke mood, establish hierarchy, and add visual interest to designs. In architecture they influence perception, atmosphere, and spatial qualities.

Texture – Texture in design refers to the tactile quality of surfaces, adding depth and visual interest to compositions.

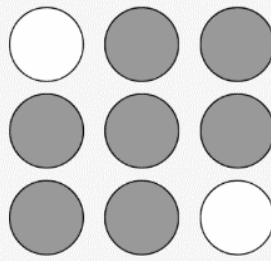
Reference: <https://archimash.com/articles/architectural-design-theory/>

DESIGN SENSITIVITY

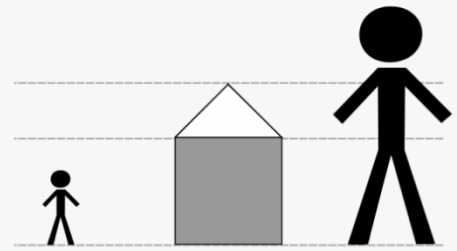
Unity/ Harmony



Balance



Proportion/ Scale



03. DESIGN THEORY contd...

Design Principles

Some basic principles of design that relate to architecture are:

Unity/Harmony – Achieving cohesion and coherence in a design by ensuring that all elements work together as a unified whole.

Balance – Achieving equilibrium in a composition through the harmonious distribution of visual elements such as colour, texture, and spatial weight.

Proportion – Establishing pleasing relationships between different parts of a whole to ensure harmony and visual appeal.

Scale – The comparative size of elements within a composition, defining their relationship to one another and the space they occupy.

Symmetry – The arrangement of elements or forms in a design that mirror each other across a central axis, creating a sense of balance and harmony.

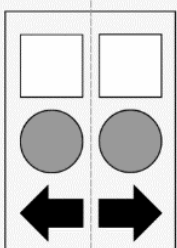
Asymmetry – The intentional deviation from a symmetrical arrangement, where elements or forms are varied in size, shape, or placement to create visual interest and dynamism within a composition.

Axis – An imaginary line or central reference point around which elements in a design are organized or balanced, influencing the spatial arrangement and symmetry of the composition.

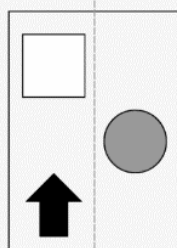
Datum – A fundamental reference plane or point within a design that establishes a baseline for spatial relationships, orientation, and organization. A datum can be horizontal, organising elements along the ground, or it can be vertical, arranging elements at a certain height. Or both.

Repetition – The deliberate use of recurring elements or motifs within a design to create visual consistency, cohesion, and a sense of wholeness.

Symmetry/ Asymmetry

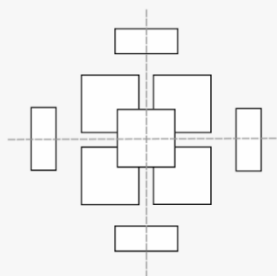


Symmetrical

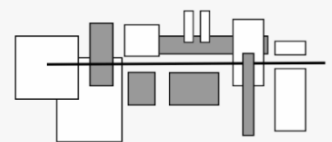


Asymmetrical

Axis/ Datum



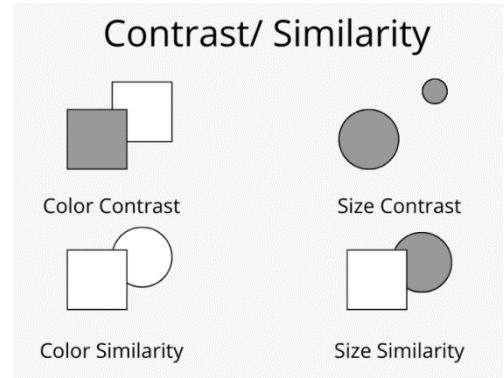
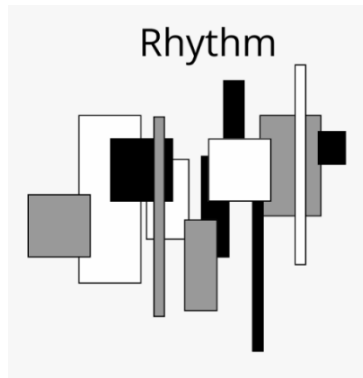
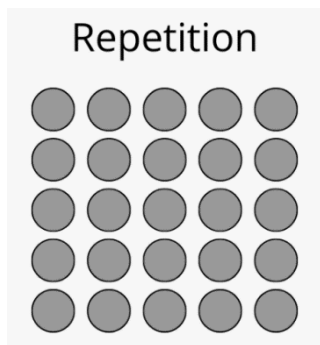
Axis



Datum

Reference: <https://archimash.com/articles/architectural-design-theory/>

DESIGN SENSITIVITY



03. DESIGN THEORY contd..

Design Principles

Repetition – The deliberate use of recurring elements or motifs within a design to create visual consistency, cohesion, and a sense of wholeness.

Rhythm – Establishing a sense of visual movement or pattern through the regular recurrence of elements or intervals within a design.

Contrast – The deliberate placement of elements with obviously different characteristics, such as color, texture, size, or shape. This creates visual interest, emphasis, and dynamic tension within a composition.

Similarity – Using shared characteristics among elements, such as color palette or shape, to establish cohesion and unity within a design.

Hierarchy – Organizing elements within a composition according to their relative importance or significance to guide the viewer's focus.

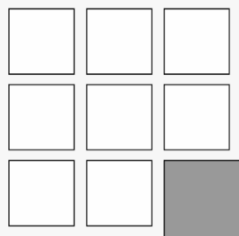
Emphasis/Dominance – Directing attention to a focal point or dominant element within a composition to create visual impact and hierarchy.

Movement – The visual sensation of motion or progression created within a design through the arrangement and interaction of elements. The viewer's eye is guided to give a dynamic sense of movement and direction.

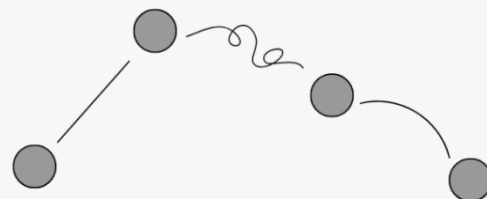
Hierarchy



Emphasis/ Dominance



Movement



Reference: <https://archimash.com/articles/architectural-design-theory/>

DESIGN SENSITIVITY

05. LOGOS

Logo design is a core tenet of graphic design—and, more specifically, of graphic design for branding. Logo designers are tasked with creating unique, memorable logos that capture the brand's identity and spark instant recognition in the viewer.

A logo is, quite simply, a visual representation of the brand. But logo design itself isn't simple! It requires a deep understanding of the brand's core values, and the ability to translate those values into a timeless yet versatile visualisation.



Think about the famous Nike “swoosh”, the iconic Google “G”, or the elaborate two-tailed siren that we immediately associate with Starbucks. All of these logos form the basis of the brands' identities—and have no doubt played an integral role in their success.

The core elements of a logo:

Most logos are built on three core elements: **Colour**, **text**, and **imagery**.

Colour

Colour plays a critical role in logo design. First and foremost, different colours can evoke different emotions and associations. At the same time, colour is essential for creating balance, harmony, and contrast—ensuring that the logo is not only aesthetically pleasing, but also easy to read and understand.

Text

The use of text in logo design varies greatly from one logo to the next. Some logos contain no text whatsoever—think of the Apple logo, which is simply a picture of an apple. Some logos comprise just one letter, like the McDonald's logo featuring the iconic “M”, while others include the full brand name, like the famous FedEx logo.

Imagery

Imagery in logo design can range from simple outlines and icons to abstract symbols, right through to elaborate illustrations—and anything else in between. Sometimes, letters are used to form imagery. And, of course, some logos don't contain any imagery at all. It all depends on what type of logo is being designed, and on the brand personality and values the logo designer wants to convey.

Reference: <https://www.andacademy.com/resources/blog/graphic-design/what-is-logo-design/>
<https://www.wix.com/blog/logo-psychology>

DESIGN SENSITIVITY

06. TYPOGRAPHY

Typography is the discipline designers use to arrange typefaces in a user interface to ensure text is legible, readable, and scalable. It must also be visually appealing to users. The right typography can enhance a product's aesthetic, optimize user-friendliness, and contribute to brand perception.

Effective typography is a blend of several elements. Here are the main ones:

Fonts and Typefaces

Fonts are specific weights within a typeface. You choose a typeface; you use a font. If you were to select Arial as your typeface, then Arial bold, italic, and regular would be your fonts. As such, a font is the distinct, stylized characteristics found within a typeface.

Letter and Line Spacing

These also go by the names of kerning and leading, respectively. They influence the readability of the text. When you optimize the space between letters and lines, you ensure the text doesn't appear too crowded or too sparse.

Calibri

Designer: Luc(as) De Groot

ABCDEFGHIJKLMNOPQRSTUVWXYZ

abcdefghijklmnopqrstuvwxyz

1234567890~!@#\$%^&*(){}'"/?

The quick brown fox jumps over the lazy dog.

The quick brown fox jumps over the lazy dog.

The quick brown fox jumps over the lazy dog.

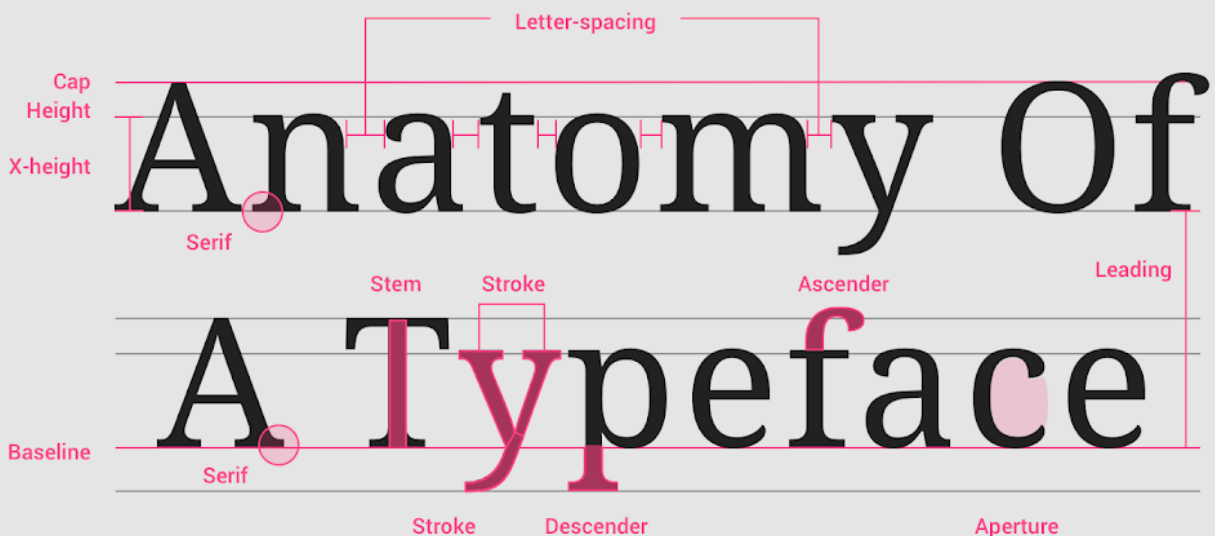
Calibri is a popular typeface with a warm and soft feel, featuring in Microsoft applications

Sand Sand

Before kerning

After kerning

Kerning in action, tightening up the letters.



Reference: <https://www.interaction-design.org/literature/topics/typography>

<https://www.interaction-design.org/literature/article/the-ux-designer-s-guide-to-typography>

DESIGN SENSITIVITY

07. MATERIALS IN DESIGN

Material is a substance or a combination of substances from which objects are made.

It is made out of tangible matter. Material Science is the study of material and their applications.

Building Materials can be classified as Natural or Man-made. <https://testbook.com/civil-engineering/building-materials-classification>

Natural Materials:

Natural materials are the materials readily available in nature. They require minimal change to be used as construction materials. Common examples of natural materials include wood, mud, stone, sand, and many more. These raw materials need no change in chemical composition. They possess all the desired qualities for construction.

BRUSH STRUCTURES

They are made of plant parts and used in primitive cultures. An extension of this is the wattle and daub process in which clay soil or dung are used to fill in and cover a woven brush structure. This gives the structure more thermal mass and strength.

Eg. Tribal Toda House



ICE AND SNOW STRUCTURES

Snow and ice were used by the Inuits for igloos and a quinzhee

A quinzhee (or quinzee) is a temporary snow shelter made from a large pile of snow which is shaped and then hollowed. In contrast, an igloo is built up from blocks of hard snow and then stacked around a circular terraced hole in the snowy ground.



MUD and CLAY structures

These are of two types:

- Walls made directly with mud mixture
- Walls built using air-dried building blocks called mud bricks.

Other uses of clay - combined with straw to make light clay, wattle and daub and mud plaster.



DESIGN SENSITIVITY

07. MATERIALS IN DESIGN contd...

SAND

Sand is used with cement to make mortar for masonry work and plaster. Sand is also part of the concrete mix.

Sandcrete block is an important low-cost material where sand is available; it is weaker but cheaper than fired clay bricks.

Sandcrete is a weak mortarlike building material with little compressive strength and zero tensile strength.



STONE and ROCK

Building stones are natural rocks or minerals that are used in construction. They are used for both structural and decorative purposes. There are many kinds of stones that can be used for building, such as basalt, marble, limestone, sandstone, quartzite, travertine, slate, gneiss, laterite, and granite.

<https://testbook.com/civil-engineering/building-stones>



GRASS

These are of many types:

- Straw
- Bamboo
- Cane
- Thatch
- Reed



Bamboo is light, strong and resilient and grows quickly, making it one of the most exciting sustainable building materials in the world. In recent years, Instead of seeing bamboo houses as primitive structures, they are considered to be at the forefront of sustainable development.

Straw bale construction is a natural building method that utilises straw as the primary building material. It typically involves tying straw into larger bundles and stacking them on top of each other in a brick-like fashion. Another common approach is to use straw bales as insulation only, and fill them into a structural frame.

Reference: <https://www.vogue.com/article/sustainable-building-bamboo-bali>



DESIGN SENSITIVITY

07. MATERIALS IN DESIGN contd...

WOOD AND TIMBER

Wood is one of the most used natural building materials in the world. Building with *wood* offers lots of advantages. *Wood* is the only *construction material* that is totally renewable, organic and climate-smart. Builders commonly use **softwood**, **hardwood**, and **engineered timber** for *construction*, as each *type* has its unique advantages and applications.



Man-Made Materials:

Natural materials are the materials that are manufactured using raw materials. They undergo physical and chemical changes and the resulting material turns out to be very different from the raw material used. Manufactured materials have greater strength and are very versatile. Common examples of these include cement, bricks, steel, glass, etc.

BRICKS

The term brick denotes a unit primarily composed of clay, but is now also used informally to denote units made of other materials or other chemically cured construction blocks. They are typically made of concrete, sand, lime, or clay. They are used to build, walls and other elements of building construction.



5 Types of Materials Used for Bricks



CONCRETE

Concrete is the most commonly used construction material in the industry. Concrete a thermal mass material; instead of it consisting of one material, it is a mix of fine aggregate, cement, water, gravel, and stones. These materials act as the foundation of many structures as well as in superstructure construction.



Common types of *concrete* used in *construction* include *plain concrete*, *precast concrete*, *ready mix concrete*, and *stamped concrete*.

Reference: <https://esub.com/blog/how-is-concrete-used-in-construction-projects/>

DESIGN SENSITIVITY

07. MATERIALS IN DESIGN contd...

STEEL

Construction is one of the most important steel-using industries, accounting for more than 50% of world steel demand. Structural steel is a popular construction material, often used alongside concrete to create impressive and long-lasting buildings. Its versatility, sustainability and flexibility are some of the main reasons for its use, as well as being a very cost-effective material.

Steel structures resist mold and mildew and are easy to maintain. One of the advantages of steel in construction is that steel buildings face fewer issues with deterioration and corrosion than concrete or wood.

Steel is also used on roofs and as cladding for exterior walls.



ALUMINIUM

Aluminum is another popular metal in modern construction.

While pure aluminum isn't suitable for building because it's not strong enough, aluminum alloys meet the requirements for construction. The most common aluminum alloys are made with copper, magnesium, and zinc. These alloys can later be cast, forged, or welded easily. So, because it's strong, lightweight, and flexible metal, it's widely used in construction for:

- Window frames, moldings, and exterior wall panels
- Roofs and walls
- Staircases and railings
- Anti-corrosive environments

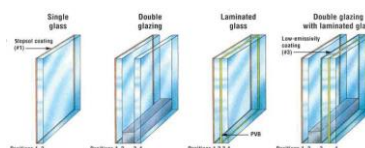


GLASS

There is a reason glass is mostly used as a building product for windows, facades, and skylights.

While it is a hard substance made from mixtures of sand, lime, and silicates, it is very brittle. On the other hand, properties such as **transparency and weather resistance** make it excellent for letting in light but protecting against the natural elements.

Most importantly, special types of glass can resist ultraviolet and infrared transmission. It is also a great insulator against heat, electricity, and electromagnetic radiation.



Reference: <https://www.constructiontuts.com/materials-used-in-building-construction/>
<https://www.planradar.com/gb/top-10-new-construction-materials-to-look-out-for-in-2024/>
<https://www.barbuliannodesign.com/post/eco-friendly-building-materials-list>

DESIGN SENSITIVITY

08. ARCHITECTURAL AWARENESS

Throughout history, architecture has been a reflection of the civilizations – their society, values, economies, culture and eventual survival/obliteration. It stands as a representation of how a community sees itself and is seen years after it had existed. The urban fabric mirrors the way people live their lives and in turn, has a deep impact on them. One cannot but wonder when we see the perfect symmetry of the temples or the technological brilliance of the skyscrapers. There is a ongoing dialogue between the built environment and its surroundings, irrespective of when it originated.

Architecture is not just limited to buildings, urban design, interiors, or landscape. It is a continuous dialogue between various subjects ranging from science and psychology, economics and sociology to documentation and history. The building cannot be perceived as just an object but is a manifestation of all the forces that has resulted in its creation be it society, economics, culture, history or technology.

The sensitivity to architecture and related forces that gave it form may be termed as architectural awareness.

History of architecture is the study of the evolution of architecture through the ages and across a wide range of geographies and cultural contexts. From the Mesopotamian civilization to the Egyptians and Greeks, architectural history is a transnational one. One of the most intriguing features of architecture is its ability to mirror the spirit of time in a way that may be even more significant than how we see it happen with art. In order to justify the parallelism between architectural history and human history, there is no better physical evidence of society change than architecture. It is possible to learn about the history of architecture by simply looking at the structures that were built in different locations at different times. Human actions were reflected in architecture, and this was made clearer by the constant effort to preserve some of the constructed history, while deciding to let the rest fade and ruin.

Tests on architectural awareness may have questions based on the following areas:

History of Buildings and its architecture.

Timeline of styles.

Building elements and style.

Materials and its applications.

Famous buildings/monuments in India

Historical Events and Related Buildings

Buildings and their Architects.

Indian Cities and Colonial history.

Famous Architectural Personalities.

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